

SHUBHAM PANCHAL

AI Student & Jr Android Dev

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SUMMARY

Motivated Android developer and junior Python programmer with hands-on experience building custom AOSP-based ROMs and developing supporting tools. Skilled in Android Studio, Material Design principles, and creating standalone Android applications such as game hubs and UI enhancement modules. Experienced in working with LineageOS/AOSP sources, OTA integration, repo syncing, and build-error parsing automation using Python. Strong understanding of software design, UML concepts, and modern development practices. Adept at problem-solving, writing clean code, and delivering fast, smooth user experiences. Passionate about mobile systems, open-source development, and building practical tools that improve workflow and performance.

TECHNICAL SKILLS

Languages: Java, Kotlin (beginner to intermediate), Python, Bash
Android Development: Android Studio, AOSP, LineageOS, custom ROM features, Material3/Material You, standalone feature apps (e.g., Game Hub), OTA integration
Python Development: CLI tools, automation scripts, repo synchronization, log parsing, build-error detection
Version Control: Git, Gerrit, GitHub
Build Systems: AOSP build system, Gradle
Tools & Technologies: Linux, VS Code, Git Bash, Shell scripting, JSON/YAML, UML (Use Case, Class, Sequence, Activity diagrams)
Areas of Focus: Performance optimization, UI modernization, ROM feature modularization

PROFESSIONAL EXPERIENCE

Android Developer | Custom ROM & App Development Jan 2023 - Present

Android Developer | Custom ROM & App Development
Developed custom AOSP/LineageOS-based ROMs with focus on smooth performance and modern UI elements.
Built standalone Android apps such as a Game Hub using Android Studio with Material 3 design.
Integrated new ROM features (QS panel enhancements, UI modifications, wallpaper modules) and prepared them for OTA releases.
Debugged AOSP builds, resolved missing blob issues, fixed dependency conflicts, and improved build reliability.
Worked with Git/Gerrit for patching, cherry-picking commits, and managing ROM source changes

Python Developer | Automation & Tools

Jan 2024 - Present

Created command-line automation tools for syncing repos, triggering ROM builds, and parsing build logs.
Designed scripts to detect missing files and errors in AOSP builds, improving productivity and reducing debugging time.
Implemented workflow automation for university projects involving ROM compilation and system setup.
Built small utilities to streamline development tasks and manage repeated processes.

Software Development Projects (Academic + Personal) Jan 2024 - Present

Designed UML diagrams (Use Case, Class, Sequence, Activity) for projects such as Student Attendance Management System and Android utilities.
Built a Python-based mini project for automating source pulling, syncing, building, and analyzing errors for custom ROMs.
Applied SDLC principles, problem-solving techniques, and clean coding practices in multiple academic assignments.

PROJECTS

1. Custom Android ROM – Velox / The Clover Project

- Built a custom AOSP/LineageOS-based ROM with focus on smoother performance and modern UI enhancements.
- Added features like refined Quick Settings, improved Settings UI, and custom wallpapers.
- Worked with source syncing, patching, cherry-picking commits, OTA packaging, and fixing build issues.
- Debugged missing blobs, dependency mismatches, and compilation errors during AOSP builds.

2. Clover Game Hub (Standalone Android App)

- Developed a dedicated gaming hub app in Android Studio using Material 3 design.
- Designed UI/UX layouts, navigation flows, and feature modules.
- Prepared the app for future integration into the ROM's ecosystem.
- Tested as a standalone APK before merging into the system image.

3. Python Build Automation Tool for Custom ROM Development

- Built a Python CLI tool to automate syncing sources, building ROMs, and reviewing build logs.
- Implemented log parsing to detect missing proprietary files and common error patterns automatically.
- Increased build workflow speed and reduced manual debugging time.
- Created clean, modular scripts suitable for ongoing ROM development

4. Student Attendance Management System (Academic Project)

- Designed complete UML diagrams including Use Case, Class, Sequence, and Activity diagrams.
- Modeled system architecture for attendance tracking, user roles, and data flow.
- Demonstrated understanding of system design and documentation best practices.

5. Web-Based Application using Python + Django (Academic Project)

- Developed a web app using Python with Django (Postgrl SQL + Django integration).
- Created basic CRUD operations and backend workflows.
- Implemented simple front-end interaction and data handling.

EDUCATION

Bachelor of Science (B.Sc) in Artificial Intelligence & Machine Learning

2nd Year – Ongoing

2023 – Present

CGPA (1st Year): 9.2

Higher Secondary Education (10+2)

Completed in 2022

ACHIEVEMENTS

- **Built two custom AOSP-based ROMs with fully working UI enhancements.**
- **Developed a standalone Android Game Hub app later prepared for ROM integration.**
- **Created a Python automation tool that reduced ROM build debugging time significantly.**
- **Maintained stable performance in academics with a 9.20 CGPA in the first year.**
- **Completed multiple university software projects involving UML, web apps, and automation.**

STRENGTHS

- Strong problem-solving mindset
- Curious and self-driven learner
- Ability to quickly adopt new frameworks and tools
- Clean and structured approach to coding
- Good balance between practical building and theoretical understanding
- Persistence in debugging and resolving complex issues

INTERESTS

- Android system development and UI/UX design
- Custom ROM engineering
- Automation scripting in Python
- AI/ML exploration and practical model implementation
- Poetry and creative writing
- Building small tools and experimental apps
- Open-source contributions

LANGUAGES (HUMAN LANGUAGES)

- English
- Hindi
- Japanese N5 (Learning)